Geometry & Trigonometry

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

A circle with centre O has radius $r = 8$ cm. Chord AB subtends a central angle $\angle AOB = \frac{3\pi}{4}$ radians. (a) Find the exact length of the minor arc AB. (b) Find the exact area of the sector OAB. (c) Find the exact area of the minor segment enclosed between chord AB and arc AB.

(a) Arc length: $s = r\theta = 8 \times \frac{3\pi}{4} = 6\pi$ cm. (b) Sector area: $A_{\text{sector}} = \frac{1}{2}r^2\theta = \frac{1}{2}(8^2)\left(\frac{3\pi}{4}\right) = 32 \times \frac{3\pi}{4} = 24\pi$ cm^2. (c) Triangle area: $A_{\text{triangle}} = \frac{1}{2}r^2\sin\left(\frac{3\pi}{4}\right) = \frac{1}{2}(64)\left(\frac{\sqrt{2}}{2}\right) = 16\sqrt{2}$ cm^2. Segment area.

2 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Consider the function $f(x) = e^{2x} - 4e^x + 3$. Let $u = e^x$. Find the minimum value of the function in terms of u.

Let the given function be $g(u) = u^2 - 4u + 3$, where $u = e^x$. Since $e^x > 0$, the domain for u is $u > 0$. To find the minimum value of the quadratic function $g(u)$, we can find the vertex. The u-coordinate of the vertex is given by $-\frac{b}{2a} = -\frac{-4}{2(1)} = 2$. Since $u = 2$ is within the domain $u > 0$, the minimum value occurs at $u = 2$. Substituting $u = 2$ into $g(u)$, we get $g(2) = (2)^2 - 4(2) + 3 = 4 - 8 + 3 = -1$. Therefore, the minimum value of the function in terms of u is -1.

3 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

The first term of a geometric sequence is $u_1 = 18$ and the common ratio is $r = \frac{1}{3}$. (a) Find the exact value of the 5th term u_5. (b) Calculate the sum of the first 6 terms, S_6. (c) State why the sum to infinity S_∞ exists, and find its exact value.

(a) $u_5 = u_1 r^4 = 18\left(\frac{1}{3}\right)^4 = 18\left(\frac{1}{81}\right) = \frac{18}{81} = \frac{2}{9}$. (b) $S_6 = \frac{u_1(1-r^6)}{1-r} = \frac{18(1-(1/3)^6)}{1-1/3} = \frac{18(1-1/729)}{2/3} = 27\left(\frac{728}{729}\right) = \frac{728}{27} \approx 26.96$. (c) Since $|r| = \left|\frac{1}{3}\right| < 1$, the terms converge to 0 and the infinite series converges. $S_\infty = \frac{u_1}{1-r} = \frac{18}{1-1/3} = \frac{18}{2/3} = 27$.

4 **[Foundational]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Group A has 20 data values with mean 15. Group B has 18 data values with mean 10. Find the mean of the combined data set.

Working through this statistics step: Combined mean $= \frac{n_1\bar{x}_1 + n_2\bar{x}_2}{n_1 + n_2} = \frac{(20)(15) + (18)(10)}{20 + 18} = \frac{240}{19}$. Final Answer: $\frac{240}{19}$

© ClipSAT — clipsat.org | All Rights Reserved

5 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Points $A(1, 5)$ and $B(-2, 0)$ are given. Find the gradient of the perpendicular bisector of $[AB]$.

Working through this modelling step: Gradient of AB: $\frac{-5}{-3}$. Perpendicular gradient $= -\frac{1}{\text{grad}_{AB}} = -\frac{3}{5}$. Final Answer: $-\frac{3}{5}$

6 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The area of a sector is 3 times its arc length for a circle of radius r. Find r.

Area $= 12r^2$, arc length $= r\theta$. Setting area $= 3 \times$ arc length: $12r^2 = 3r12r = 3r = 6$ (dividing both sides by $r\theta eq0$).

7 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 6x + 8$ in applications of exponential functions, evaluate $f(5)$.

Final Answer: 38. Substitute $x = 5$: $f(5) = 6(5) + 8 = 38$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

For the model $f(x) = 2x + 9$ in the natural exponent, evaluate $f(2)$.

Final Answer: 13. Substitute $x = 2$: $f(2) = 2(2) + 9 = 13$.

9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A data set recorded for the natural exponent consists of the four values $\{12, 16, 20, 26\}$. What is the arithmetic mean? Show your complete algebraic work.

Final Answer: 18.5. The sum is $12 + 16 + 20 + 26 = 74$, and dividing by 4 gives 18.5.

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Practice 3.1: Solve the equation $2\cos 2x = 4\cos x - 3$ for $0 \le x \le 2\pi$, giving exact solutions.

1. Substitute the double angle identity $\cos 2x = 2\cos^2 x - 1$:

$2(2\cos^2 x - 1) = 4\cos x - 3 \Rightarrow 4\cos^2 x - 2 = 4\cos x - 3 \Rightarrow 4\cos^2 x - 4\cos x + 1 = 0.$ 2. Recognize the perfect square:

$(2\cos x - 1)^2 = 0 \Rightarrow 2\cos x - 1 = 0 \Rightarrow \cos x = \frac{1}{2}$. 3. Reference angle: $\alpha = \arccos(1/2) = \frac{\pi}{3}$. Cosine is positive in Quadrants I and IV: In Q1: $x = \frac{\pi}{3}$. In Q4: $x = 2\pi - \frac{\pi}{3} = \frac{5\pi}{3}$. Exact solutions.

© ClipSAT — clipsat.org | All Rights Reserved

11 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)**

Practice 3.2: Ship A leaves port P on a bearing of 040° traveling at 15 km/h. At the same time, ship B leaves P on a bearing of 130° traveling at 20 km/h. Find the distance between the two ships after 2 hours.

1. Distance travelled by ship A: $d_A = 15 \times 2 = 30$ km. 2. Distance travelled by ship B: $d_B = 20 \times 2 = 40$ km. 3. The angle between their paths is the difference in bearings: $\theta = 130^\circ - 40^\circ = 90^\circ$. 4. Since the triangle is right-angled at P, apply the Pythagorean theorem (or Cosine Rule with $\cos 90^\circ = 0$): $d = \sqrt{30^2 + 40^2} = \sqrt{900 + 1600} = \sqrt{2500} = 50$ km. The distance between the ships after 2 hours is exactly 50 km.

12 **[Intermediate]** • **Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)**

[Geometry & Trigonometry · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

Substitute $t = 8$: $g(8) = 5(8) + 5 = \mathbf{45}$. Setting $5t + 5 = 50$ gives $5t = 45 \Rightarrow t = \mathbf{9}$.

13 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)**

Practice AI.1: Use the trapezoidal rule with 4 subintervals to estimate $\int_1^3 \sqrt{1 + x^3}\, dx$.

1. Subinterval width: $h = \frac{b-a}{n} = \frac{3-1}{4} = 0.5$. 2. Evaluation grid points: $x_0 = 1, x_1 = 1.5, x_2 = 2.0, x_3 = 2.5, x_4 = 3.0$. 3. Compute function values $y_i = \sqrt{1 + x_i^3}$: $y_0 = \sqrt{1+1} = \sqrt{2} \approx 1.4142$ $y_1 = \sqrt{1 + 1.5^3} = \sqrt{1 + 3.375} = \sqrt{4.375} \approx 2.0917$ $y_2 = \sqrt{1 + 2^3} = \sqrt{9} = 3.0000$ $y_3 = \sqrt{1 + 2.5^3} = \sqrt{1 + 15.625} = \sqrt{16.625} \approx 4.0774$ $y_4 = \sqrt{1 + 3^3} = \sqrt{28} \approx 5.2915$ 4. Apply the trapezoidal r.

14 **[Intermediate]** • **Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)**

Practice 1.1: In an arithmetic sequence, the 4th term is 19 and the sum of the first 10 terms is 290. Find the first term u_1, the common difference d, and the sum of terms from u_{11} to u_{25} inclusive.

1. $u_4 = u_1 + 3d = 19$. 2. $S_{10} = \frac{10}{2}(2u_1 + 9d) = 5(2u_1 + 9d) = 290 \Rightarrow 2u_1 + 9d = 58$. 3. Multiply the first equation by 2: $2u_1 + 6d = 38$. Subtract from the second: $3d = 20 \Rightarrow d = 6$ (checking: $2u_1 + 54 = 58 \Rightarrow 2u_1 = 4 \Rightarrow u_1 = 2$, and $u_4 = 2 + 3(6) = 20 eq 19$; let us solve exact: $3d = 20 \Rightarrow d = \frac{20}{3}$, $u_1 = 19 - 20 = -1$). If $u_4 = 19$ and $S_{10} = 235$: $2u_1 + 9d = 47$, $2u_1 + 6d = 38 \Rightarrow 3d = 9 \Rightarrow d = 3$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

© ClipSAT — clipsat.org | All Rights Reserved

15 **[Challenging]** • **Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)** • **Cross-Lesson: 2.01 Functions**

Practice 5.2: A particle moves in a straight line with velocity $v(t) = 3t^2 - 12t + 9$ m/s for $0 \le t \le 4$. (a) Find the times at which the particle is momentarily at rest. (b) Calculate the net displacement of the particle over $[0, 4]$. (c) Calculate the total distance travelled by the particle over $[0, 4]$.

(a) $v(t) = 0 \Rightarrow 3(t^2 - 4t + 3) = 0 \Rightarrow 3(t-1)(t-3) = 0 \Rightarrow t = 1$ s and $t = 3$ s. (b) Displacement: $\Delta s = \int_0^4 (3t^2 - 12t + 9)\,dt = [t^3 - 6t^2 + 9t]_0^4 = (64 - 96 + 36) - 0 = 4$ m. (c) Total distance requires splitting the integral where velocity changes sign: Over $[0, 1]$: $v(t) > 0$; $\int_0^1 (3t^2 - 12t + 9)\,dt = [t^3 - 6t^2 + 9t]_0^1 = 4$ m. Over $[1, 3]$: $v(t)0$.

16 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)** • **Cross-Lesson: 4.01 Statistics & Probability**

Practice 5.1: A curve has equation $y = x^3 - 3x^2 - 9x + 5$. (a) Find the coordinates of the local maximum and local minimum points. (b) Determine the coordinates of the point of inflection and find the equation of the tangent at the point of inflection.

(a) $y' = 3x^2 - 6x - 9 = 3(x^2 - 2x - 3) = 3(x-3)(x+1)$. Critical points at $x = -1$ and $x = 3$. $y'' = 6x - 6$. At $x = -1$: $y''(-1) = -120 \Rightarrow$ local minimum. $y(3) = 27 - 27 - 27 + 5 = -22$. Local min: $(3, -22)$. (b) Point of inflection: $y'' = 6x - 6 = 0 \Rightarrow x = 1$. $y(1) = 1 - 3 - 9 + 5 = -6$. Inflection point: $(1, -6)$. Slope at $x = 1$: $m = y'(1) = 3(1)^2 - 6(1) - 9 = -12$. Tangent line: $y - (-6) = -12(x - 1) \Rightarrow y + 6 = -12x + 12 \Rightarrow y = -12x + 6$.

17 **[Challenging]** • **Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)** • **Cross-Lesson: 5.01 Calculus**

Practice AI.2: A company takes out a loan of 120,000 to purchase equipment at an annual interest rate of 6% compounded monthly, repayable by equal monthly payments over 10 years. (a) Find the amount of each monthly payment. (b) Calculate the total interest paid over the life of the loan.

(a) Use the annuity formula or TVM solver: Number of payments $N = 10 \times 12 = 120$. Monthly interest rate $r = \frac{0.06}{12} = 0.005$. $PV = PMT\left[\frac{1-(1+r)^{-N}}{r}\right] \Rightarrow 120,000 = PMT\left[\frac{1-(1.005)^{-120}}{0.005}\right]$. Compute discount factor: $1 - (1.005)^{-120} = 1 - 0.549633 = 0.450367$. Factor / 0.005 $= 90.07345$. $x1{,}332.25$. $(b) Total amount paid$: 120 \times 1332.25 = \159, 870. Total interest: $x159{,}870$ - \$39,870.

© ClipSAT — clipsat.org | All Rights Reserved

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 AI SL — Specific Topics

Practice 1.2: An investor deposits 8,000 into a high-yield account earning an annual nominal interest rate of 4.8% compounded monthly. Determine the account balance after 6 years, and find the number of full years required for the investment to double in value.

1. $FV = PV\left(1 + \frac{r}{100k}\right)^{kn}$. Here $PV = 8000$, $r = 4.8$, $k = 12$, $n = 6$: $x10{,}660.80$.
2. $For the investment to double$: 8000(1.004)^{12n} = 16000 \implies (1.004)^{12n} = 2
. $Take natural logs$: 12n \ln(1.004) = \ln(2) \implies n = \frac{\ln 2}{12 \ln(1.004)} \approx \frac{0.693147}{12(0.003992)} \approx \frac{0.693147}{0.0479} \approx 14.47 years. Therefore, 15 full years are required for the balance to double.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number & Algebra

Practice 2.2: The graph of $y = f(x)$ undergoes three consecutive transformations: (1) a horizontal stretch by scale factor 3, (2) a reflection in the y-axis, and (3) a translation by vector $(4;\ -5)$. Write down the equation of the resulting function $g(x)$ in terms of f.

1. Horizontal stretch by factor 3: $x \to \frac{x}{3}$, giving $y = f\left(\frac{x}{3}\right)$. 2. Reflection in y-axis: $x \to -x$, giving $y = f\left(-\frac{x}{3}\right)$. 3. Translation by $(4;\ -5)$: replace x by $x - 4$ and subtract 5: $g(x) = f\left(-\frac{x-4}{3}\right) - 5 = f\left(\frac{4-x}{3}\right) - 5$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Functions

An IB AA SL student scores 64/80 on Paper 1 (40%), 72/80 on Paper 2 (40%), and 16/20 on the Internal Assessment (20%). Calculate the student's overall weighted percentage.

Paper 1: $\frac{64}{80} = 80\%$. Paper 2: $\frac{72}{80} = 90\%$. IA: $\frac{16}{20} = 80\%$. Overall weighted percentage: $0.40(80) + 0.40(90) + 0.20(80) = 32 + 36 + 16 = 84\%$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 4x + 6$ in rational exponents, evaluate $f(2)$.

a. 14 ✓

b. 18

c. 8

d. 18

Substitute $x = 2$: $f(2) = 4(2) + 6 = 14$.

© ClipSAT — clipsat.org | All Rights Reserved

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

For the model $f(x) = 5x + 7$ in exponential functions, evaluate $f(2)$.

a. 17 ✓

b. 22

c. 10

d. 21

Substitute $x = 2$: $f(2) = 5(2) + 7 = 17$.

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for rational exponents consists of the four values $\{18, 22, 26, 32\}$. What is the arithmetic mean?

a. 22.5

b. 23.5

c. 24.5 ✓

d. 26.5

The sum is $18 + 22 + 26 + 32 = 98$, and dividing by 4 gives 24.5.

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A data set recorded for exponential functions consists of the four values $\{10, 14, 18, 24\}$. What is the arithmetic mean?

a. 14.5

b. 15.5

c. 16.5 ✓

d. 18.5

The sum is $10 + 14 + 18 + 24 = 66$, and dividing by 4 gives 16.5.

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for applications of exponential functions consists of the four values $\{11, 15, 19, 25\}$. What is the arithmetic mean?

a. 15.5

b. 16.5

c. 17.5 ✓

d. 19.5

The sum is $11 + 15 + 19 + 25 = 70$, and dividing by 4 gives 17.5.

© ClipSAT — clipsat.org | All Rights Reserved

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for logarithms consists of the four values $\{13, 17, 21, 27\}$. What is the arithmetic mean?

a. 17.5

b. 18.5

c. 19.5 ✓

d. 21.5

The sum is $13 + 17 + 21 + 27 = 78$, and dividing by 4 gives 19.5.

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 3x + 3$ in logarithms, evaluate $f(3)$.

a. 12 ✓

b. 15

c. 9

d. 16

Substitute $x = 3$: $f(3) = 3(3) + 3 = 12$.

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

A data set recorded for logarithm laws and change of base consists of the four values $\{14, 18, 22, 28\}$. What is the arithmetic mean?

a. 18.5

b. 19.5

c. 20.5 ✓

d. 22.5

The sum is $14 + 18 + 22 + 28 = 82$, and dividing by 4 gives 20.5.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

For the model $f(x) = 4x + 4$ in logarithm laws and change of base, evaluate $f(3)$.

a. 16 ✓

b. 20

c. 12

d. 20

Substitute $x = 3$: $f(3) = 4(3) + 4 = 16$.

© ClipSAT — clipsat.org | All Rights Reserved

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

In the diagram, triangle ABC is shown with coordinates A(2, 1), B(7, 1), and C(5, 5). What is the length of the side AC?

a. 4

b. 5

c. $2\sqrt{5}$ ✓

d. $\sqrt{20}$

To find the length of the side AC, we use the distance formula between points A(1, 2) and C(3, 6). The distance formula is given by $d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$. Substituting the coordinates of A and C: $AC = \sqrt{(3-1)^2 + (6-2)^2}$ $AC = \sqrt{(2)^2 + (4)^2}$ $AC = \sqrt{4 + 16}$ $AC = \sqrt{20}$ To simplify $\sqrt{20}$, we find the largest perfect square factor of 20, which is 4. $\sqrt{20} = \sqrt{4 \times 5} = \sqrt{4} \times \sqrt{5} = 2\sqrt{5}$. Therefore, the length of side AC is.

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

In the right-angled triangle ABC shown below, the length of side AB is 10 units and the length of side BC is 6 units. The angle ACB is 90 degrees. Find the length of side AC.

a. 10

b. 6

c. **8** ✓

d. 16

The triangle ABC is a right-angled triangle with the right angle at C. We can use the Pythagorean theorem, which states that in a right-angled triangle, the square of the hypotenuse (the side opposite the right angle) is equal to the sum of the squares of the other two sides. In this case, AB is the hypotenuse. So, we have: $AB^2 = AC^2 + BC^2$. We are given $AB = 10$ and $BC = 6$. We need to find AC. Substituting the given values into the equation: $10^2 = AC^2 + 6^2$ $100 = AC^2 + 36$ $AC^2 = 100 - 36$.

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Volume of cone with radius 3, height 4:

a. 12π ✓

b. 16π

c. 36π

d. 48π

Applying Geometry & Trigonometry principles, $V = \frac{1}{3}\pi(9)(4) = 12\pi$. Therefore, the verified correct option is 12π.

© ClipSAT — clipsat.org | All Rights Reserved

33 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In triangle ABC, $A = 39^\circ$, $B = 39^\circ$, and $a = 13$. Find b, correct to 1 decimal place.

a. **13.0** ✓

b. 15.0

c. 14.0

d. 11.0

$\frac{a}{\sin A} = \frac{b}{\sin B} \Rightarrow b = \frac{13\sin(39^\circ)}{\sin(39^\circ)} \approx 13.0$.

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

A right triangle has legs 12 and 16 and hypotenuse 20. The angle θ is opposite the side of length 12. Find $\cos(\theta)$.

a. $\frac{4}{3}$

b. $\frac{3}{5}$

c. $\frac{3}{4}$

d. $\frac{4}{5}$ ✓

$\cos(\theta) = \frac{\text{adjacent}}{\text{hypotenuse}} = \frac{16}{20} = \frac{4}{5}$.

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

A sector of a circle with radius 6 cm has area 27 cm^2. Find the angle of the sector, in radians.

a. **1.5** ✓

b. 3

c. 0.75

d. 2

$27 = \frac{1}{2}(6^2)\theta = 18\theta \Rightarrow \theta = 1.5$ rad.

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

A right triangle has legs of length 18 and 24. Find the length of the hypotenuse.

a. 32

b. 34

c. 33

d. **30** ✓

By Pythagoras: $18^2 + 24^2 = 30^2$, so the unknown side is 30.

© ClipSAT — clipsat.org | All Rights Reserved

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

A circle has radius 8 cm. Find the arc length subtended by an angle of 1.5 radians.

a. **12 cm ✓**

b. 6 cm

c. 24 cm

d. 9.6 cm

Arc length $= r\theta = 8 \times 1.5 = 12$ cm.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 6.01 AI SL — Specific Topics

In the diagram, points A, B, and C form a triangle. Point D lies on AC such that BD is the angle bisector of angle ABC. Given that AB = 10 cm, AC = 15 cm, and BC = 12 cm, and that AD = 6 cm, find the length of DC.

a. 8 cm

b. **7.2 cm ✓**

c. 9 cm

d. 6 cm

According to the Angle Bisector Theorem, an angle bisector of a triangle divides the opposite side into two segments that are proportional to the other two sides of the triangle. In triangle ABC, BD is the angle bisector of angle ABC. Therefore, we have the proportion: $\frac{AD}{DC} = \frac{AB}{BC}$ We are given that AB = 10 cm, AC = 15 cm, and BC = 12 cm. We are also given that AD = 6 cm. Let DC = x cm. Substituting the given values into the proportion: $\frac{6}{x} = \frac{10}{12}$ To solve for x, we can cross-m.

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.01 Number & Algebra

In the given right-angled triangle ABC, angle B is 90 degrees. Point D is on BC such that BD = 4 and DC = 9. If AD is the altitude to the hypotenuse AC, find the length of AD.

a. 6 ✓

b. 8

c. 12

d. 15

In a right-angled triangle, the altitude to the hypotenuse creates two smaller triangles that are similar to the original triangle and to each other. \\ According to the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. \\ In triangle ABC, AD is the altitude to the hypotenuse BC. Therefore, we have: \\ $AD^2 = BD \times DC$ \\ Given that BD = 4 and DC = 9: \\ $AD^2 = 4 \times 9$ \\ $AD^2 = 36$ \\.

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 2.01 Functions

In the diagram, triangle ABC has angle ABC equal to 90 degrees. Point D lies on AC such that BD is perpendicular to AC. If AD = 4 cm and DC = 9 cm, find the length of BD.

a. 5 cm

b. **6 cm** ✓

c. 7 cm

d. 3 cm

In a right-angled triangle, when an altitude is drawn from the right angle to the hypotenuse, it creates two smaller triangles that are similar to the original triangle and to each other. This leads to the geometric mean theorem. The length of the altitude (BD) is the geometric mean of the two segments it divides the hypotenuse into (AD and DC). Therefore, $BD^2 = AD \times DC$. Given AD = 4 cm and DC = 9 cm, $BD^2 = 4 \times 9$ $BD^2 = 36$ $BD = \sqrt{36}$ $BD = 6$ cm.

© ClipSAT — clipsat.org | All Rights Reserved

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.01 Statistics & Probability

In the given triangle ABC, AB = 10, AC = 15, and BD = 6. D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. **9** ✓

b. 10

c. 15

d. 22.5

By the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments created by the bisector on the opposite side. Therefore, $\frac{AB}{AC} = \frac{BD}{DC}$. Substituting the given values, we get $\frac{10}{15} = \frac{6}{DC}$. Simplifying the left side, $\frac{2}{3} = \frac{6}{DC}$. Cross-multiplying, $2 \times DC = 3 \times 6$. $2 \times DC = 18$. Dividing by 2, $DC = \frac{18}{2} = 9$.

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Calculus

In the right-angled triangle ABC, shown below, AD is the altitude to the hypotenuse BC. If BD = 4 and DC = 9, find the length of the altitude AD.

a. 5

b. **6** ✓

c. 7

d. 8

In a right-angled triangle, the altitude to the hypotenuse creates similar triangles. By the Geometric Mean Theorem, the square of the altitude to the hypotenuse is equal to the product of the lengths of the two segments it divides the hypotenuse into. Thus, $AD^2 = BD \times DC$. Given $BD = 4$ and $DC = 9$, we have $AD^2 = 4 \times 9 = 36$. Taking the square root of both sides, we get $AD = \sqrt{36} = 6$.

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 6.01 AI SL — Specific Topics

In the triangle ABC, AB = 10, AC = 15, BD = 6, and D is a point on BC such that AD is the angle bisector of angle BAC. Find the length of DC.

a. 7

b. **9** ✓

c. 10

d. 12

According to the Angle Bisector Theorem, the ratio of the sides adjacent to the angle bisector is equal to the ratio of the segments it divides the opposite side into. Therefore, $\frac{AB}{AC} = \frac{BD}{DC}$. Substituting the given values, we have $\frac{10}{15} = \frac{6}{x}$, where x is the length of DC. Solving for x: $10x = 15 \times 6$ $10x = 90$ $x = \frac{90}{10}$ $x = 9$.

© ClipSAT — clipsat.org | All Rights Reserved

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.01 Number & Algebra

A sector of a circle has radius 9 cm and angle 45°. Find the exact area of the sector.

a. $\frac{81\pi}{8}$ ✓

b. $\frac{49\pi}{8}$

c. 8π

d. $\frac{9\pi}{2}$

$A = \frac{1}{2}r^2\theta = \frac{1}{2}(9)^2\left(\frac{1\pi}{4}\right) = \frac{81\pi}{8}$ cm^2.

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 2.01 Functions

In triangle ABC, $a = 5$, $b = 10$, and $C = 120^\circ$. Find the exact length of c.

a. $5\sqrt{7}$ ✓

b. $\sqrt{179}$

c. 15

d. $3\sqrt{19}$

$c^2 = a^2 + b^2 - 2ab\cos C = (5)^2 + (10)^2 - 2(5)(10)\cos(120^\circ) = 175 \Rightarrow c = 5\sqrt{7}$.

© ClipSAT — clipsat.org | All Rights Reserved